



CONTACT

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LOCATION
Campinas, SP, Brazil

LINKEDIN
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SKILLS

- C/C++ARM Cortex-MFreeRTOS
- Embedded LinuxCAN/CANopen
- SPI/I2C/UARTJTAG/SWD Debugging
- Oscilloscope & Logic Analyzer
- RTOS-based Firmware Architecture

LANGUAGES

- PortugueseNative
- EnglishFluent
- SpanishConversational

CERTIFICATIONS

ISTQB Certified Tester - Foundation Level

Rafael Augusto Almeida Costa

Senior Embedded Systems Engineer

PROFILE

Senior Embedded Systems Engineer with 12+ years of experience designing, integrating, and validating firmware and hardware for industrial, automotive, and IoT products. Strong background in real-time systems, low-power embedded design, communication protocols, and cross-functional delivery from prototype to production.

EXPERIENCE

Senior Embedded Systems Engineer · Bosch BrasilJan 2021 - Present

- Led embedded firmware development for connected sensor modules, reducing release defects by 38% over three product cycles.
- Defined board bring-up and validation procedures for 6+ hardware revisions, cutting prototype debug time by 30%.
- Coordinated with hardware, test, and manufacturing teams to launch 4 production programs on schedule.
- Improved boot and communication stack performance by 22% through task optimization and protocol tuning.

Embedded Software Architect · WEG AutomaçãoAug 2017 - Dec 2020

- Architected firmware for industrial control devices using FreeRTOS and C++, supporting over 50,000 deployed units.
- Standardized driver and middleware layers across product lines, reducing code duplication by 45%.
- Implemented CAN and Modbus integration features that expanded interoperability with legacy equipment.
- Mentored a team of 5 engineers and established code review and regression testing practices.

Embedded Systems Engineer · EmbraerMar 2013 - Jul 2017

- Developed and validated embedded control software for avionics support equipment, meeting strict traceability requirements.
- Built automated hardware-in-the-loop test setups that increased test coverage by 28%.
- Worked on real-time communication and fault-handling logic, improving system resilience during stress testing.
- Supported certification documentation and design reviews across engineering and quality teams.

Firmware Engineer · Positivo TecnologiaFeb 2010 - Feb 2013

- Created low-level firmware for consumer electronics products, including peripheral drivers and startup code.
- Reduced firmware memory footprint by 18% through stack optimization and module refactoring.
- Supported PCB bring-up, signal debugging, and production testing for multiple high-volume releases.

EDUCATION

M.S. in Electrical Engineering
Universidade Estadual de Campinas (UNICAMP) · 2012

B.S. in Computer Engineering
Universidade de São Paulo (USP) · 2009